PHYSICS · HSSC PART I · CHAPTER 1

Physical Quantities and Measurement

Dimensions · Precision & Accuracy · Uncertainty

SLO 1.2.3 — SLO 1.4.2

Today's Learning Objectives

By the end of this period, you will be able to:



1.2.3 – 1.2.4

Dimensions

Show the homogeneity of physical equations and derive formulas for physical quantities using dimensional analysis.



1.3.1 – 1.3.2

Precision & Accuracy

Differentiate between precision and accuracy, and analyse data collected using different measuring instruments.



1.4.1 – 1.4.2

Significant Figures & Uncertainty

Solve word problems on uncertainty with correct significant figures, and justify why every measurement carries uncertainty.

Quick Recap: Dimensions

Every physical quantity can be expressed in terms of the seven base dimensions.

Base Quantity	Dimension	SI Unit
Length	**[L]**	metre (m)
Mass	**[M]**	kilogram (kg)
Time	**[T]**	second (s)
Electric Current	**[A]**	ampere (A)
Temperature	**[K]**	kelvin (K)
Amount of Substance	**[mol]**	mole (mol)
Luminous Intensity	**[cd]**	candela (cd)



Why it matters

Dimensions strip away numbers and units, leaving only the underlying physical nature of a quantity — e.g. velocity is always $[LT^{-1}]$, no matter what unit system you use.

This lets us:

- Check if an equation is dimensionally consistent
- Derive formulas connecting physical quantities

The Principle of Homogeneity

SLO 1.2.3 — showing the homogeneity of physical equations

An equation is dimensionally homogeneous only if the dimensions of every term on the left-hand side match the dimensions of every term on the right-hand side.

1

Identify the terms

List every term appearing in the equation, separated by + , − , or =.

2

Find each dimension

Replace every physical quantity with its dimensional formula [M, L, T, ...].

3

Compare both sides

Only terms added or subtracted — or the two sides of '=' — must carry identical dimensions.

Key idea: *you cannot add a length to a time — homogeneity is what makes that visible.*

Worked Example — Checking Homogeneity

Verify: $s = ut + \frac{1}{2}at^2$ *(equation of motion)*

LHS	$\mathbf{s}$	$= [L]$
Term 1 (RHS)	$\mathbf{ut}$	$= [LT^{-1}] \times [T] = [L]$
Term 2 (RHS)	$\mathbf{\frac{1}{2}\,at^2}$	$= [LT^{-2}] \times [T^2] = [L]$ (½ is a pure number, dimensionless)

Conclusion: every term reduces to [L]. LHS = RHS in dimensions, so the equation is homogeneous — and therefore dimensionally valid.

Deriving Formulas Using Dimensions

SLO 1.2.4 — dimensional analysis as a formula-building tool

1

Propose the relation

Assume the quantity depends on others as a power product:
$Q = k \cdot A^x \cdot B^y \cdot C^z$

2

Write dimensions

Replace every quantity (both sides) with its dimensional formula [M, L, T].

3

Equate powers

Match the powers of M, L and T on both sides to get simultaneous equations.

4

Solve for x, y, z

Solve the equations, then substitute back to get the formula (k stays undetermined).

Worked Example — Period of a Simple Pendulum

Assume period T depends on length l, mass m, and acceleration due to gravity g.

$$T = k \cdot l^{x}\, m^{y}\, g^{z}$$

$$[T] = [L]^{x}\,[M]^{y}\,[LT^{-2}]^{z}$$

$$[T^{1}] = [M]^{y}\,[L]^{x+z}\,[T]^{-2z}$$

Compare powers:

M: $0 = y \rightarrow y = 0$

T: $1 = -2z \rightarrow z = -\tfrac{1}{2}$

L: $0 = x + z \rightarrow x = \tfrac{1}{2}$

Substitute back

$x = \tfrac{1}{2}, \quad y = 0, \quad z = -\tfrac{1}{2}$

$$T = k \cdot l^{1/2} \cdot g^{-1/2}$$

$$T = k\sqrt{(l / g)}$$

Dimensional analysis alone cannot find k — experiment shows k = 2π. Note that mass drops out entirely (y = 0): the period of a pendulum does not depend on its mass.

Precision vs. Accuracy

SLO 1.3.1 — two different, and easily confused, qualities of measurement

Accuracy

How close a measurement is to the true (accepted) value.

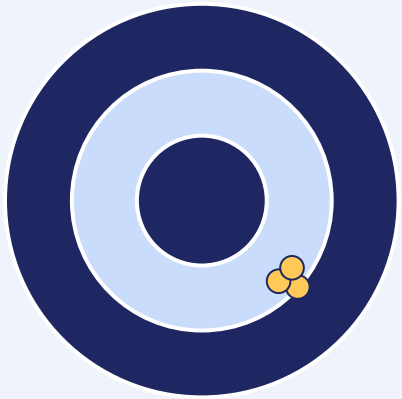
Precision

How close repeated measurements are to one another, regardless of the true value.



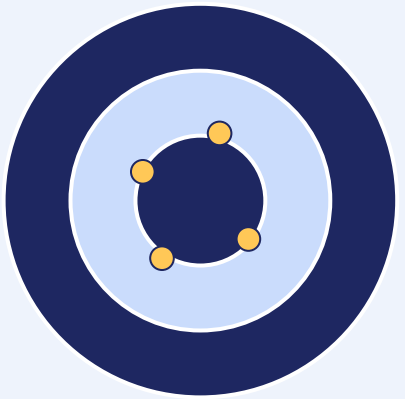
Accurate & Precise

hits clustered on centre



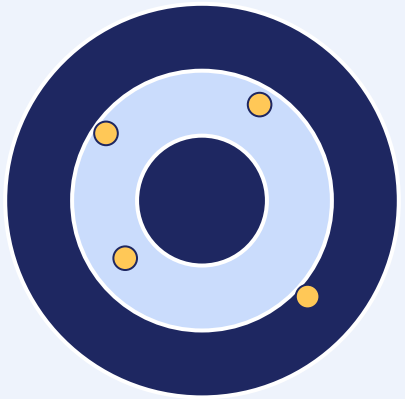
Precise, not Accurate

tight cluster, off-centre



Accurate, not Precise

scattered around centre



Neither

scattered, off-centre

Precision Across Instruments

SLO 1.3.2 — analysing data collected with different measuring instruments

Instrument	Least Count (Precision)	Typical Use
Metre Rule	**± 0.1 cm (1 mm)**	Length of a table, distance travelled
Vernier Calliper	**± 0.01 cm**	Diameter of a test tube, small rod
Screw Gauge (Micrometer)	**± 0.001 cm**	Diameter of a thin wire
Digital Stopwatch	**± 0.01 s**	Time for a pendulum oscillation

Finer least count → higher precision. A screw gauge gives more precise readings than a metre rule — but an instrument can still be inaccurate if it is poorly calibrated or wrongly zeroed, no matter how precise it is.

Uncertainty in Calculated Quantities

SLO 1.4.1 — carrying the correct number of significant figures through a calculation

Addition / Subtraction

Result keeps as many decimal places as the term with the fewest decimal places.

$12.63\ \text{cm} + 3.1\ \text{cm} = 15.73\ \text{cm} \rightarrow 15.7\ \text{cm}$

Multiplication / Division

Result keeps the same number of significant figures as the term with the fewest.

$4.5\ \text{cm} \times 2.10\ \text{cm} = 9.45\ \text{cm}^2 \rightarrow 9.5\ \text{cm}^2$

Worked word problem

A rectangular plate measures 15.2 cm by 8.24 cm. Find its area to the correct number of significant figures.

$\text{Area} = 15.2 \times 8.24 = 125.248\ \text{cm}^2.$ 15.2 has 3 s.f. (fewest) → round to 3 s.f. → Area = 125 cm²

Why Every Measurement Has Uncertainty

SLO 1.4.2 — justifying that no measurement is ever perfectly exact



Instrument Limit

Every instrument has a least count — a smallest division it can resolve. Anything finer is estimated.



Estimation by Eye

Reading between the smallest divisions (or a needle position) always involves human judgement.



Environmental Factors

Temperature, vibration, air currents, and parallax subtly shift readings from trial to trial.



The Quantity Itself

Some quantities (like a rough or irregular surface) don't have one single 'true' value to find.

Conclusion: a measurement is only meaningful when reported as a best value ± its uncertainty — never as a single ‘exact’ number.

Key Takeaways

1. A homogeneous equation has matching dimensions on every term — this both checks and helps derive formulas.

2. Accuracy is closeness to the true value; precision is closeness among repeated readings — they are independent qualities.

3. Different instruments have different least counts, so their precision differs even when measuring the same quantity.

4. Calculated results must be rounded to the correct significant figures, set by the least precise measurement used.

5. Every measurement carries some uncertainty, from the instrument, the observer, or the environment — report it as best value ± uncertainty.